Simulating Gravitational Differences: The Free Fall of a 20-Peso Coin Across Solar System Planets at Heights of 50m, 80m, and 100m

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COLLEGE PHYSICS 1

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Abstract:
This project explores the concept of free fall across different planets in our solar system using common objects such as 20-peso coins. The study applies Galileo’s foundational principle that free fall acceleration remains constant for any object under uniform gravitational force, while acknowledging that gravity varies depending on the mass and radius of the planet. The simulation calculates the time to fall and final velocity for the coin from a height of 100 meters, 80 meters, and 50 meters on Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto. By integrating equations of motion, the results highlight the influence of planetary gravity, such as Jupiter's high acceleration yielding shorter fall times. Findings are compared with standard gravitational data, confirming the simulation's accuracy and demonstrating the difference of free fall in a visual way. This provides insights into how gravity impacts motion across different planetary environments.

Introduction:

Free fall is a fundamental concept in physics, describing the motion of an object under the influence of gravity alone, with no other forces, such as air resistance, acting on it. In a state of free fall, objects experience constant acceleration, which is determined by the gravitational force of the planet or celestial body they are on. The acceleration due to gravity is often denoted by g, and it varies significantly depending on the mass of the planet.

The Earth's gravitational acceleration is approximately 9.81 m/s which serves as a standard reference for many studies involving free fall and gravitational effects. However, gravitational acceleration is not uniform across all planets in the Solar System.

The difference in gravitational forces significantly impacts the dynamics of free-falling of common objects such as coins. On planets with higher gravitational accelerations, such as Jupiter and Neptune, objects experience shorter fall times and higher final velocities. Conversely, on planets with weaker gravitational fields, such as Mercury and Pluto, fall times are longer, and final velocities are lower.

The gravitational accelerations for selected planets in our Solar System are provided in Physica Cloud (2017)

Planet	$Gravity = g\ (m/s^2)$
Mercury	$3.7\ m/s^2$
Venus	$8.87\ m/s^2$
Earth	$9.81\ m/s^2$
Mars	$3.71\ m/s^2$

Jupiter	$10.44\ m/s^2$
Saturn	$10.44\ m/s^2$
Uranus	$8.69\ m/s^2$
Neptune	$11.15\ m/s^2$
Pluto:	$0.62\ m/s^2$

Problem Statement

How does gravitational acceleration influence the fall time and velocity of a common object like a coin on different planets?

Objectives

1. To simulate the free fall of a common object, such as a 20-peso coin on various planets.
2. To compute fall times, final velocities, and accelerations using equations of motion at heights of 50m, 80m, and 100m.
3. To visualize and analyze the differences in free-fall behavior based on planetary gravity.

Scope and Limitations

This study models free fall under ideal conditions, ignoring air resistance and rotational effects. It focuses on nine planets in our solar system.

LITERATURE REVIEW

Free Fall and Earth's Gravity

Summarize existing studies or projects relevant to your topic: The NASA Beginner's Guide to Aeronautics explains that free fall occurs when the only force acting on an object is gravity. On Earth, this results in a consistent acceleration of approximately 9.8 m/s^2, known as the gravitational constant (NASA, n.d.). Galileo Galilei's groundbreaking experiments laid the foundation for understanding free fall by demonstrating that objects fall at the same rate in a vacuum, regardless of their mass.

Highlight gaps your project intends to address: While existing studies focus on the behavior of free fall on Earth, they often lack detailed exploration of how gravitational variations across different celestial bodies influence free-fall acceleration. This study addresses this gap by simulating the motion of a 20-peso coin on planets within the solar system, providing a comparative perspective on gravitational effects.

Discuss key theories, principles, or frameworks guiding your work: This research is guided by the principles of kinematic motion. Specifically, the equations of motion for free-falling objects are used to analyze the coin's behavior under different gravitational accelerations.

The primary formula applied is:

where is the final velocity, is the initial velocity (zero in free-fall scenarios), is the acceleration due to gravity, and is the time of fall.

The formula for displacement:

is used to calculate the distance traveled during free fall. These kinematic equations provide a robust framework for modeling the motion of a 20-peso coin in free fall across various planetary environments.

Gravitational Variations Across Celestial Bodies

Summarize existing studies or projects relevant to your topic: The "Free-Fall Acceleration" study on Physica Cloud discusses the universal formula for calculating gravitational acceleration on celestial bodies:

where it represents the gravitational constant, the celestial body's mass, and its radius (Physica Cloud, 2017). The study highlights variations in gravitational pull across planets and moons, influenced by differences in their physical characteristics. For

example, Jupiter's gravity is significantly stronger than Earth's, while the Moon's is much weaker.

Highlight gaps your project intends to address: While these calculations are well-documented, the practical simulation of free-fall scenarios for everyday objects, such as a coin, across multiple planets is less explored. This project aims to fill this void by applying theoretical knowledge to a tangible simulation, enhancing understanding of gravitational differences in a relatable context.

Discuss key theories, principles, or frameworks guiding your work: The study employs kinematic equations of motion to model free-fall scenarios under varying gravitational conditions. This study calculates the final velocity and displacement of the 20-peso coin for each planetary environment. The resulting simulation demonstrates how gravitational forces influence the motion of objects in diverse planetary conditions, offering insights into their real-world implications.

Methodology:

Research Design/Approach

The project applies computational simulation using Python (Pygame) to model and Matplotlib for data visualization.

Tools and Materials

1. **Programming Language**: Python
2. **Libraries**: Pygame, Matplotlib

Procedure

1. **Define Gravity Constants**: Values were set for each planet based on known gravitational accelerations.
2. **Python was used to program equations for fall time and velocity:**
    - A Python script calculated fall time and final velocity.

Fall Time

Kinematic formula for displacement under constant acceleration:

$$Y_f - Y_o = V_0 t + \frac{1}{2} g t^2$$

The Initial velocity (V_0) is zero, the formula simplifies to:

$$Y = \frac{1}{2} g t^2$$

Solving for T gives:

$$Fall\ Time\ (t):\ \ t = \sqrt{\frac{2Y}{g}}$$

Vf: Final velocity of the object.

V0: Initial velocity of the object.

g: Acceleration due to gravity.

t: Time of fall.

Final Velocity

Kinematic formula with constant acceleration.

$$Final\ Velocity\ V_f = V_0 - gt$$

The initial velocity (V_0) is zero, the formula simplifies to:

$$Final\ Velocity\ (V_f):\ V_f = gt$$

3. **Pygame was used to animate coins dropping from a fixed height.**
4. **Data Visualization**: Matplotlib generated bar graphs comparing fall times and velocities across planets.

Data Collection and Analysis

Simulated data included fall times and final velocities. Results were organized and visualized using bar graphs for comparison.

Results and Discussions

The simulation of free-fall motion of common objects such as coins across different planets provided insights into the effects of varying gravitational accelerations on fall time and final velocity.

The results were compiled for each of the planets in the simulation, and the fall times and final velocities were recorded. These results can be seen in the table below:

Results for Height: 50m

Planet	V0 (m/s)	G (m/s²)	t (s)	Vf (m/s)
Jupiter	0	24.79	2.01	49.79
Neptune	0	11.15	2.99	33.39
Saturn	0	10.44	3.09	32.31
Earth	0	9.81	3.19	31.32
Venus	0	8.87	3.36	29.78
Uranus	0	8.69	3.39	29.48
Mars	0	3.71	5.19	19.26
Mercury	0	3.7	5.20	19.24
Pluto	0	0.62	12.70	7.87

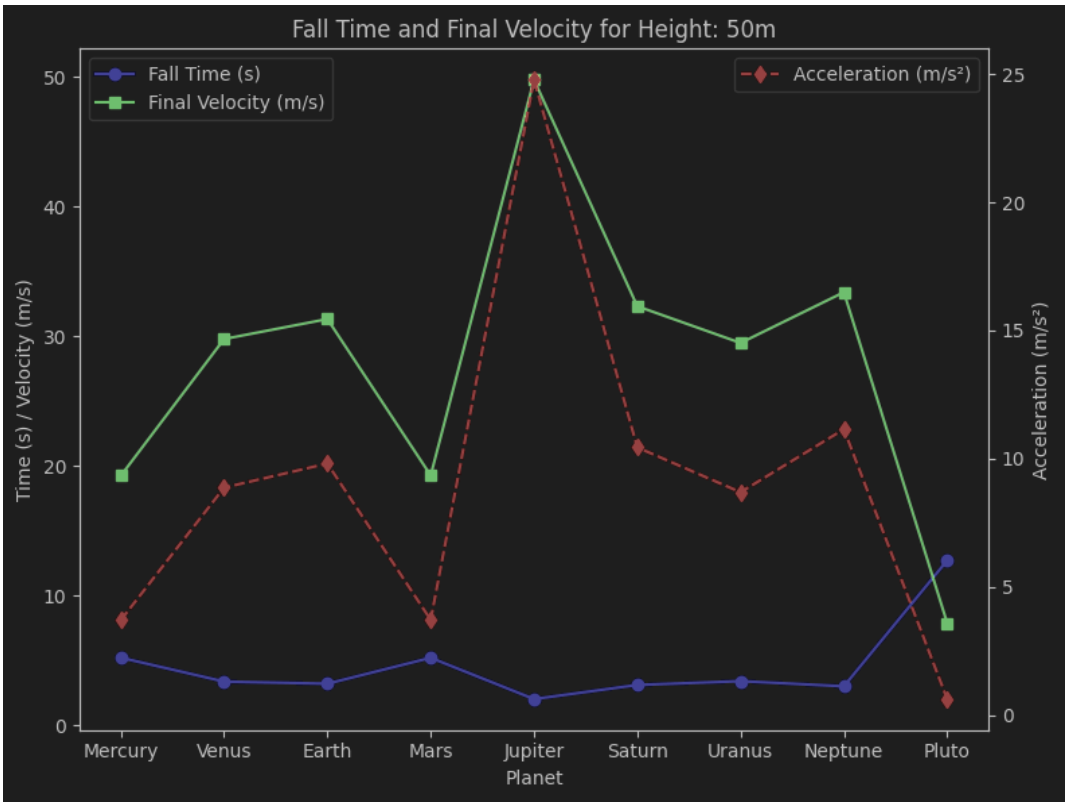
Results for Height: 80m

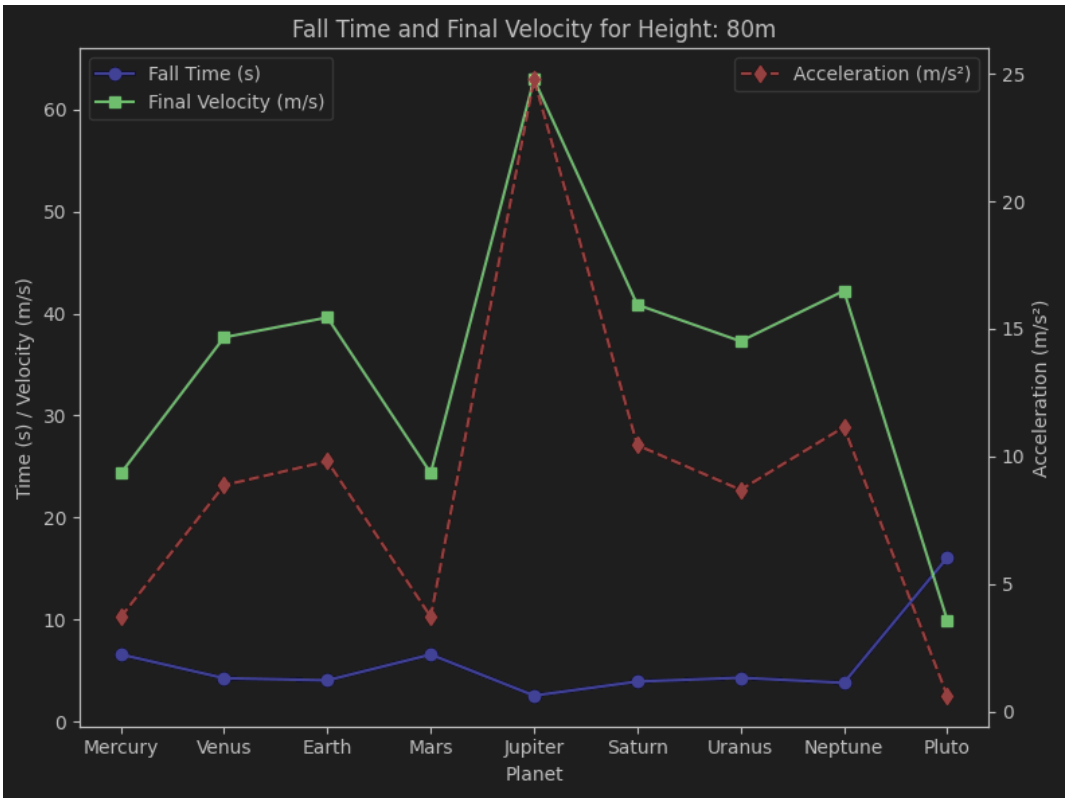
Planet	V0 (m/s)	G (m/s²)	t (s)	Vf (m/s)
Jupiter	0	24.79	2.54	62.98
Neptune	0	11.15	3.79	42.24
Saturn	0	10.44	3.91	40.87
Earth	0	9.81	4.04	39.62
Venus	0	8.87	4.25	37.67
Uranus	0	8.69	4.29	37.29
Mars	0	3.71	6.57	24.36
Mercury	0	3.7	6.58	24.33
Pluto	0	0.62	16.06	9.96

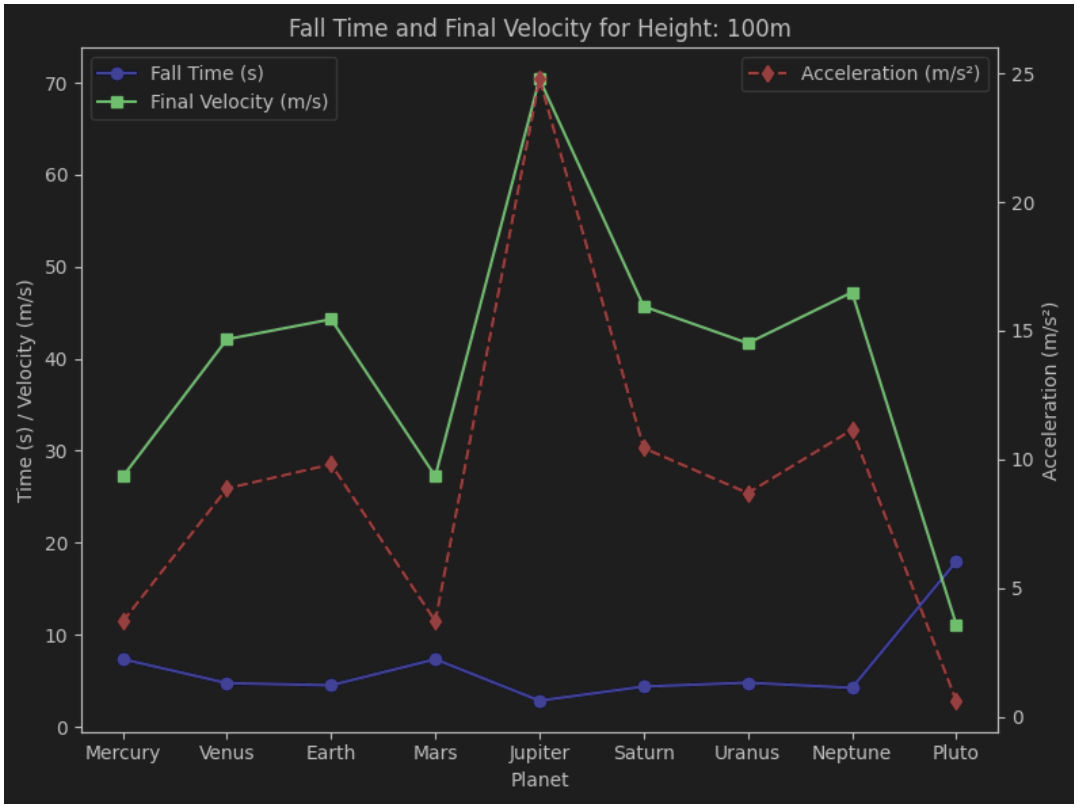
Results for Height: 100m

Planet	V0 (m/s)	G (m/s²)	t (s)	Vf (m/s)
Jupiter	0	24.79	2.84	70.41
Neptune	0	11.15	4.24	47.22

Saturn	**0**	**10.44**	**4.38**	**45.69**
Earth	**0**	**9.81**	**4.52**	**44.29**
Venus	**0**	**8.87**	**4.75**	**42.12**
Uranus	**0**	**8.69**	**4.80**	**41.69**
Mars	**0**	**3.71**	**7.34**	**27.24**
Mercury	**0**	**3.7**	**7.35**	**27.20**
Pluto	**0**	**0.62**	**17.96**	**11.14**




Fall Time and Final Velocity for Height: 80m
Fall Time (s)
Final Velocity (m/s)
Acceleration (m/s²)
Time (s) / Velocity (m/s)
Acceleration (m/s²)
0
10
20
30
40
50
60
0
5
10
15
20
25
Mercury
Venus
Earth
Mars
Jupiter
Saturn
Uranus
Neptune
Pluto
Planet



Insights from Graphs

Findings for Fall Time and Final Velocity

The simulation investigates the gravitational differences by analyzing the free fall of a 20-peso coin across the planets in the solar system. Results are presented for three heights: 50m, 80m, and 100m. **Fall time (t)** and **final velocity (Vf)** show clear variations depending on gravitational acceleration (G), which differs for each planet.

Height: 50m

- Jupiter has the shortest fall time of **2.01 seconds** and the highest final velocity of **49.79 m/s** due to its high gravitational acceleration (**24.79 m/s²**).
- Pluto exhibits the longest fall time (**12.70 seconds**) and the lowest final velocity (**7.87 m/s**) because of its minimal gravitational acceleration (**0.62 m/s²**).
- Earth serves as a middle reference point with **3.19 seconds** fall time and **31.32 m/s** velocity.

Height: 80m

- The trends remain consistent: Jupiter has the fastest fall (**2.54 seconds**) and highest final velocity (**62.98 m/s**), while Pluto has the slowest fall (**16.06 seconds**) and lowest final velocity (**9.96 m/s**).
- Earth again lies mid-range with a fall time of **4.04 seconds** and a velocity of **39.62 m/s**.

Height: 100m

- Similar trends persist with Jupiter leading in speed (**2.84 seconds**, **70.41 m/s**) and Pluto trailing with a time of **17.96 seconds** and velocity of **11.14 m/s**.
- Earth's values are **4.52 seconds** for time and **44.29 m/s** for velocity.

Acceleration and its Influence

- **Gravitational Acceleration (G):** The acceleration determines the rate of change of velocity during free fall. Higher values of **G** (e.g., Jupiter's **24.79 m/s²**) lead to shorter fall times and higher final velocities.
- **Acceleration Trends:**
    - Jupiter's acceleration dominates, resulting in the fastest free fall.
    - Mars and Mercury have lower accelerations (**3.71 m/s²** and **3.7 m/s²**, respectively), leading to longer fall times.
    - Pluto's gravitational pull is minimal (**0.62 m/s²**), producing the slowest fall and lowest velocity.

Visual Trends

From the graphs:

- **Fall time** decreases as gravitational acceleration increases.
- **Final velocity** increases proportionally with gravitational strength.
- Neptune, Saturn, Uranus, and Venus follow a similar middle range, while Jupiter and Pluto represent the extremes.

Challenges and Implications

- The gravitational variations highlight the challenges of object motion across planets with diverse gravitational forces neglecting air resistance
- Findings can be applied in space exploration, understanding gravity-assisted motion, and calibrating planetary simulations

Interpretation of Results

The findings from the simulation align with the fundamental principles of Newtonian mechanics, specifically the relationship between gravitational acceleration and the motion of falling objects. By comparing fall times and final velocities across different planetary bodies

Understanding these differences can inform landing strategies for spacecraft on various planets, where both the fall time and final velocity must be considered to ensure safe landing operations.

Implications, Challenges, and Contributions

This study provides a valuable tool for visualizing and comparing the effects of different gravitational accelerations on free-fall motion of common objects such as coins. It can be used in educational contexts to help students grasp the concept of gravity and free fall across various celestial bodies.

However, there are challenges and limitations to the simulation:

- Simplification of Factors: The simulation assumes a vacuum and neglects the effects of air resistance, which would significantly impact fall dynamics, especially on planets with atmospheres like Earth and Venus.
- Assumptions of Constant Gravity: The simulation assumes a constant gravitational force for each planet, which isn't the case in reality due to variations in gravity with altitude

Unexpected Results and Limitations

One unexpected result was the nearly identical fall times and final velocities observed for Mercury and Mars, despite differences in their size and mass. This similarity arises because their gravitational accelerations are nearly the same (**3.7 m/s²** for Mercury and **3.71 m/s²** for Mars).

Another key observation was Pluto's drastically longer fall times and significantly lower final velocities compared to other planets. This highlights the pronounced effect of weak gravity (**0.62 m/s²**) at the edge of our Solar System, contrasting sharply with Jupiter's strong gravitational pull (**24.79 m/s²**), which produces the fastest falls.

The study's limitations include:

1. **Exclusion of Atmospheric Resistance:** The results assume a vacuum environment. Planets with atmospheres, like Earth and Venus, would show different fall times and velocities if air resistance were considered.

2. **Constant Gravitational Acceleration:** The gravitational forces were simplified as constant values, which may not reflect reality, especially for large planets like Jupiter where gravity varies with altitude.

Conclusion

Summary of Findings

This study successfully demonstrated the variations in free-fall behavior across different planets by simulating the motion of a **20-peso coin** dropped from heights of **50m**, **80m**, and **100m**. The key findings include:

- **Jupiter** (gravitational acceleration: **24.79 m/s²**) produces the **shortest fall times** and **highest final velocities**. For example:
  - At **50m**: Fall time = **2.01s**, Final velocity = **49.79 m/s**.
  - At **100m**: Fall time = **2.84s**, Final velocity = **70.41 m/s**.
- **Pluto** (gravitational acceleration: **0.62 m/s²**) exhibits **significantly longer fall times** and **lower velocities**, showing weak gravity:
  - At **50m**: Fall time = **12.70s**, Final velocity = **7.87 m/s**.
  - At **100m**: Fall time = **17.96s**, Final velocity = **11.14 m/s**.
- **Mercury** and **Mars**, despite differences in size and mass, have similar fall times and velocities due to their nearly identical gravitational accelerations (**3.7 m/s²** and **3.71 m/s²**, respectively).

Achievement of Objectives

The project successfully met its objectives by:

1. Simulating the free fall of a coin on **nine planets** across three heights (50m, 80m, and 100m).
2. Calculating fall times, final velocities, and accelerations using the **equations of motion**.
3. Visualizing the results through **graphs** and **tables**, highlighting significant trends.

Recommendations for Future Work

To enhance and expand this study, the following recommendations are proposed:

1. **Include Atmospheric Effects:** Account for air resistance on planets with significant atmospheres (e.g., Earth, Venus, and Jupiter) to produce more realistic fall times and velocities.
2. **Analyze Rotational Forces:** Investigate how planetary rotation and centrifugal forces influence free-fall dynamics, particularly on rapidly rotating planets like Jupiter and Saturn.
3. **Expand the Scope:** Explore free-fall behavior on **moons**, **dwarf planets**, or **exoplanets** to broaden the study's applications in space exploration and planetary science.

References

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